



ITC — Information Technology Colleague

Digital Egypt Pioneers Initiative - DEPI



MATAMAK

Restaurant Management System

Web-Based • ASP.NET Core 8 • Angular 17

Full Stack .NET

DEPI — Team 1

2025 / 2026

SHR4_SWD5_S1

Full Stack .NET Track (Sharkia)

Special thanks to the initiative's management and to ITC, the company implementing the Full Stack .NET track (Sharkia).



Overview

AGENDA

What we'll cover today

01 Introduction

02 Problem Statement

03 Solution Overview

04 System Design

05 Demo & Screenshots

06 Conclusion

MATAMAK is a comprehensive web-based restaurant management system built with ASP.NET Core 8 & Angular 17. It covers the full operations lifecycle — from customer ordering and payment processing to admin analytics and inventory control.

87%

Test Coverage

100%

UAT Completion

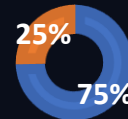
4.7

Satisfaction /5

6/6

KPIs Met

■ Passed ■ Failed





Project Overview

Team · Tech Stack · Supervisor



The Team Members

M

Mostafa Mahmoud Amin

Team Leader & Full Stack Developer

Z

Zyad Ayman Abdel-Salam

Full Stack Developer

Project Name

MATAMAK — Restaurant Management System

A

Ahmed Mohamed Abdullah

System Analyst

E

Ibrahim Medhat Abbas

DB Administrator

Tech Stack

ASP.NET Core 8 · Angular 17 · SQL Server · EF Core · JWT

B

Belal Muhammad Abdo

Frontend Developer

Supervisor

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Problem Statement

Most restaurants in Egypt rely on manual systems or outdated, standalone software, leading to order errors, slow billing, and a lack of comprehensive analytical insight.

01 Order tracking errors

Orders get lost or duplicated due to the lack of a centralized digital record, slowing down the kitchen and harming the customer experience.

02 Poor inventory management

Staff discover a shortage of ingredients only at the time of preparation, causing delays and embarrassment in front of customers.

03 Absence of analyses

Without organized data collection, the restaurant owner cannot see the best-selling items or peak times.



SOLUTION

Proposed Solution



MATAMAK — An integrated web-based restaurant management platform that brings all operations together in one place.

The main solution

A comprehensive web-based system that manages the entire order lifecycle: from menu browsing to invoice generation and analytics.

Advantages

Role-based access (3 roles) · Order tracking · Admin analytics dashboard

Technologies used

ASP.NET Core 8 · Angular 17 · SQL Server 2022 · EF Core · JWT · Angular Material

Expected results

API response under 0.8 seconds · 100% request accuracy · 4.7/5 user satisfaction



System Design

1. Three-Tier Architecture
2. Components
3. RESTful API — HTTPS/JSON



Frontend

Angular 17 · TypeScript · Angular
Material · RxJS · Lazy Loading



Backend

ASP.NET Core 8 · JWT Auth · RBAC · 3-
Tier Architecture · EF Core



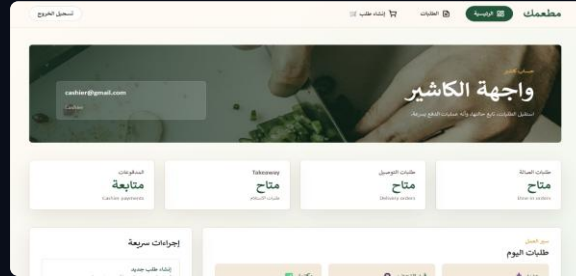
Database

SQL Server 2022 · 9 Tables · 3NF · Code-
First Migrations

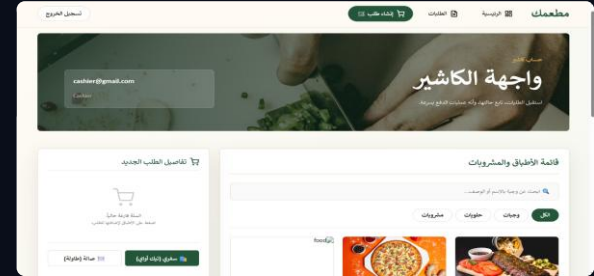
Demo & Screenshots



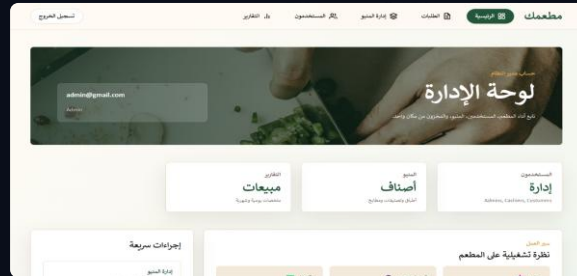
Customer Home — Menu Browsing



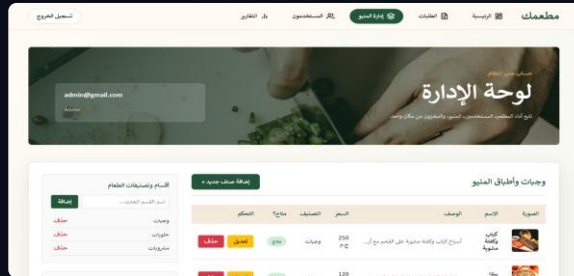
Cashier Dashboard — Overview



Cashier — Create New Order



Admin Panel — Analytics Dashboard



Admin Panel — Menu Management



Project Results

MATAMAK Team — Achieved KPIs



87%

Unit Test Coverage

100%

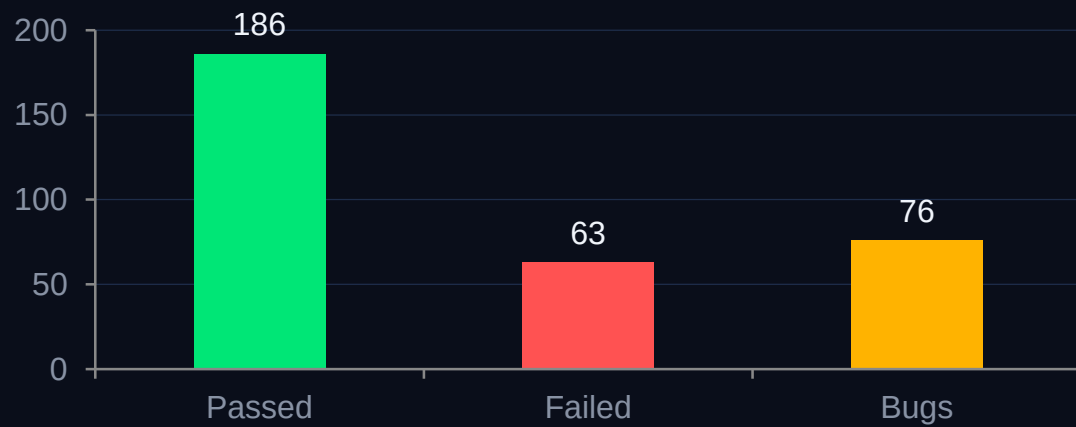
UAT Completion

4.7

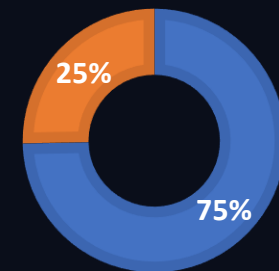
Satisfaction /5

0.8s

Avg. API Response



■ Passed (74.7%) ■ Failed (25.3%)





Conclusion

Summary

MATAMAK has been fully delivered—31 functional requirements, 7 system tests (all passed), and 87% module coverage.

Learnings

Clean Architecture in .NET · RESTful API Design · Team Management via GitHub

Future Work

Redis Caching · Docker & Azure Cloud Deployment

Thank You!

Mostafa Mahmoud Amin — Team Leader
MATAMAK · ITC Graduation Project · 2025–2026